

Adding a reusable representation beside a fixed prediction service: what coalition-conditioned erasure buys, and what it does not

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Abstract

An organisation already publishes a prediction service: fixed probability vectors that downstream recipients consume and cannot be asked to re-accept. We ask whether it can add a *second*, reusable representation for one recipient without changing a single published number, while limiting what that recipient—and a coalition of recipients—can newly infer about sex and race. The mechanism we study keeps the teacher information the published outputs do not explain and chooses directions by maximising linear reconstruction of that residual minus conditional-moment penalties, conditioning on the released service probabilities through cross-fitted nuisance models. Penalties can be local to one recipient or specific to a coalition’s joint view. We evaluate eight penalty settings against five frozen adversarially trained channels and a no-channel baseline on California ACS, then transport the entire frozen 14-interface slate to a locked, previously unused survey year. Four findings. (i) Exact service preservation is structural and holds bitwise on all 210 released views; it does not imply stable service accuracy, which moves by ± 0.014 nats across one year. (ii) On the sealed evaluation the coalition-conditioned penalty does what it claims: it lowers additional sensitive recovery against both an equal-strength and an equal-total-mass local control, under simultaneous 95% intervals and both weightings, satisfying a rule the development year had failed. (iii) That rule is a one-sided constraint on a *point* estimate, and we show the interval evidence does *not* establish non-inferiority at the same 0.001 margin—zero of four comparisons, under any procedure we computed. (iv) The mechanism nevertheless loses: under identical attacks it is significantly worse than a frozen adversarially trained channel of the same width on all four sensitive endpoints under both weightings, and we show this ordering survives matching the two channels on average utility, in both directions. We report an evaluation contribution and a negative method result, and we are explicit about which of our own earlier claims this revision withdraws.

1 What this paper is

A service that already publishes predictions can hand one recipient extra features built only from what those predictions leave unexplained, and can steer those features away from directions that would let that recipient—or that recipient plus a colluding one—guess sex or race better than the published predictions already allow. We show the steering knob works on a sealed new year of data, that it is not competitive with an adversarially trained channel of the same size, and that the negative verdict is robust to matching the two on average utility. The useful output is therefore the problem formulation and the evaluation design, plus a negative method result. It is not a deployable mechanism, and we do not present it as one.

We also correct our own record. Section 8 lists every claim this revision withdraws or narrows, with the evidence. No number changed: all 90 pre-declared endpoints reproduce under an independently written scorer to 1.1×10^{-15} .

2 Threat model and interface

Each person has covariates X , sensitive attributes S (sex, 2 classes; race, 9 classes) and task labels. Two frozen services exist: $H_A(X) \in [0, 1]^4$, the income-over-\$50k and employment probability vectors released to

recipient A, and $H_B(X) \in [0, 1]^2$, public coverage, released to recipient B. A mechanism may add a channel $Z \in \mathbb{R}^{16}$ for A *only*. Views: A sees (H_A, Z) ; B sees H_B ; the coalition AB sees (H_A, Z, H_B) .

The policy authorises A for income, employment and residence, B for coverage and commute, and forbids sex and race everywhere, coverage and commute for A, and income, employment and residence for B—eleven forbidden roles. The adversary holds labelled data from the release year, fits fixed attack families on its own pool, selects on a validation pool, and sees only its legal view. Every published function, including the preprocessing, the released map and the neural arms’ public auxiliary heads, is in scope, and coalition attacks may use any legal projection of A’s and B’s candidates.

The A-only limitation is real and bounds every claim below. Only one recipient receives a channel. Nothing here demonstrates joint optimisation of channels for several recipients, and the coalition penalty is a constraint on one channel given another recipient’s fixed view, not a multi-channel design.

3 Why the published outputs are retained

The constraint that makes this problem different from representation learning is that H_A and H_B have already been published. Downstream consumers have built on those exact numbers; a revision is a migration, not a deployment. So the mechanism may append but not modify: Z is concatenated to H_A and never mixed into it. Exact preservation is then *structural* rather than measured—there is no perturbation whose size must be bounded—and we verify it bitwise on every released view rather than certifying it statistically.

Two things this does not give us. It does not mean the service stays *accurate*: on identical released vectors, employment log loss rises by 0.0138 nats and income falls by 0.0145 between the two years (Table 5 discusses the probe side). And it does not constrain what the appended channel discloses, which is the entire subject of this paper.

4 The mechanism

Let T be a frozen PCA32 teacher. Build $\phi(T) = [\text{standardised } T, 96 \text{ random Fourier features}]$ with a median-heuristic bandwidth [Rahimi and Recht, 2007]. Let q_A be the intercept and all degree-2 monomials in A’s two positive-class service probabilities; q_{AB} adds B’s. Residualise by SVD least squares, $V_0 = \phi - q_A B_\phi$ and $R = T - q_A B_T$, then eigen-whiten V_0 to V with $V^\top V/n = I$. Utility is $U = C_{VR} C_{RV}$: for $W^\top W = I$, $\text{tr}(W^\top U W)$ is exactly the training reduction in least-squares reconstruction error of the residualised teacher. Protection matrices use household-grouped 3-fold cross-fitted multinomial-logistic nuisances m_j (cross-fitting in the sense of Chernozhukov et al., 2016; we claim none of that work’s orthogonality or rate guarantees for our moments) and, for each retained basis column b_k ,

$$G_{jk} = \frac{1}{n_j} \sum_i V_i b_k(H_i) (\text{onehot}(S_j)_i - m_j(H_i))^\top, \quad P_j = \sum_k G_{jk} G_{jk}^\top \text{ (trace-normalised),}$$

averaged over local roles (basis q_A : sex, race, coverage), coalition roles (basis q_{AB} : sex, race) and a marginal variant (constant basis). The released map is the top-16 eigenvectors of $U_{\text{norm}} - \lambda P$, giving arms S0 ($\lambda = 0$), M025/M1 (marginal), L025/L1/L2 (local) and C025/C1 (local + coalition). Inference needs only T and H_A ; H_B enters fitting and auditing only.

4.1 What is a theorem, and what is not

The eigenvector step is Ky Fan’s maximum principle for a *fixed* empirical matrix [Fan, 1949]. It certifies nothing about the choice of features, basis, nuisances, rank or attacker, and with r fixed at 16 it can retain directions whose penalty exceeds their utility. Selecting rank by eigenvalue sign instead is not our idea and is not new: it is SARL’s Theorem 3, OptNet-ARL’s Theorem 4.1 and K-TOpt’s Corollary 4.1 [Sadeghi et al., 2019, 2021, 2022]. We fixed $r = 16$ to match the width of the neural baselines, which is a comparison choice, not an optimum.

The moment condition is a finite subset of the L^2 characterisation of conditional independence that kernel CI tests use [Zhang et al., 2011, Strobl et al., 2019]. Precisely, it is Lemma 2(v) of Zhang et al. [2011], attributed there to Daudin: our centring on the S side is algebraically equivalent to residualising the (Z, H) side, since with $\tilde{f} = f - \mathbb{E}[f | H]$ we have $\mathbb{E}[f(Z, H)(\mathbf{1}\{S = c\} - m_c(H))] = \mathbb{E}[\tilde{f} \cdot \mathbf{1}\{S = c\}]$. The characterisation quantifies over *every* square-integrable f ; we test $f \in \{Z_\ell b_k(H)\}$ —linear in Z , degree-2 in H , with estimated nuisances and finite samples. Consequently vanishing fitted moments do not establish conditional independence, and a non-zero moment may reflect nuisance error rather than leakage. Two fixtures make this concrete and are carried in the code. An XOR channel has exactly zero marginal moment while $S = ZH$ recovers it perfectly; it is detected by an interaction term *in* H , so enriching the conditioning side suffices. A magnitude channel— (Z, S) uniform on $\{(-1, 0), (1, 0), (-2, 1), (2, 1)\}$ with H constant—has exactly zero first moment, yet $\mathbf{1}\{|Z| > 1.5\}$ recovers S exactly; because H is constant, *no* enrichment of the conditioning side can ever see it, and only a nonlinear function of Z can ($\mathbb{E}[Z^2(S - \frac{1}{2})] = 0.75$). The attacks that defeat our arms on ACS are MLPs and boosted trees, i.e. the second kind.

With population-optimal log-loss predictors, the H versus (H, Z) risk difference equals $I(S; Z | H)$. Our validation-selected finite attacks do not attain those infima, so the reported differences bound it in neither direction—indeed some are negative—and mutual-information estimation has its own formal limits [McAllester and Stratos, 2020]. We report no certified information quantity. For the withholding control, an independent visible branch $B \sim \text{Bernoulli}(p)$ gives $I(S; B, Z_B | H) = pI(S; Z | H)$ and expected routed loss $(1 - p)L_H + pL_{HZ}$; the branch must be persistent, since independent re-draws would eventually reveal Z .

5 Relation to prior work

The closed-form line of Sadeghi, Dehdashtian and Boddeti already solves this shape of problem. SARL [Sadeghi et al., 2019] takes eigenvectors of a utility-minus- λ -dependence matrix; OptNet-ARL [Sadeghi et al., 2021] keeps closed-form ridge players inside a learned encoder and already handles several sensitive attributes through several λ ; K-TOpt [Sadeghi et al., 2022] adds random Fourier features and evaluates on Folktables [Ding et al., 2021]; U-FaTE [Dehdashtian et al., 2024] adds a *conditional* penalty. Our marginal arms are, up to parametrisation, SARL’s construction on whitened random-feature residuals.

Three things differ, and we state their weight honestly. *What we condition on*: U-FaTE’s conditional measure is stratification over a discrete label Y , and its own text scopes the equivalence to “when Y is not a continuous label”; H_A is a pair of continuous probability vectors, so stratification is unavailable and we substitute cross-fitted nuisance residuals. H is also a *released artifact* rather than the label to be predicted—a different object even setting continuity aside. *What utility means*: reconstruction of a residualised teacher rather than dependence with a label. *Recipient structure*: separate local and coalition penalties, with only one recipient receiving a channel.

The first two are adaptations of known components, not new optimisation principles. The third we did not find in the surveyed literature, and the nearest candidate is weaker prior art than its title suggests: the multi-consumer information-theoretic line [Sankar et al., 2013] designs *one* sanitised release consumed by many users who differ only in private side information, with no per-recipient release, no collusion term, and no constraint to preserve an already-published output. Linear concept erasure [Belrose et al., 2023, Holstege et al., 2025] addresses a different guarantee (exact *linear* erasure) on a representation the releaser owns and may edit; LEACE scopes its guarantee to linear adversaries explicitly. Adversarial removal is known to leave recoverable information [Elazar and Goldberg, 2018], and least-privilege learning has fundamental limits [Stadler et al., 2024].

Two further neighbourhoods are worth separating from ours. Conditional fair representation learning conditions on a *label* for separation-style criteria [Zhao et al., 2019, Hwa et al., 2024], not on a released prediction, and again designs the representation from scratch. Information-theoretic privacy gives the quantities our log-loss endpoints relate to—leakage and utility both become mutual information under log loss [Makhdoumi et al., 2014], with perfect-privacy conditions [Rassouli and Gündüz, 2021] and robustness of leakage measures to side information [Liao et al., 2019]—but assumes a known joint distribution and a randomised mapping, whereas our released map is deterministic and our attacks are fitted. Task-agnostic private representation learning [Arevalo et al., 2024] targets attribute inference without a known downstream

task, again with no fixed prior output and one recipient. A bounded search cannot establish novelty and we make no priority claim.

6 Experimental design

California ACS one-year PUMS, ages 19–34. *Development:* 2018, three household splits (seeds), with disjoint pools for representation fitting, source validation, task fitting and validation, attacker fitting and validation, and development evaluation. Fourteen interfaces per seed—H (no channel), E, A0, L025, L20, J (frozen adversarially trained channels, recovered without retraining) and the eight spectral arms—i.e. 42 interface instances, of which 24 are the new maps, frozen before any reserved-label evaluation.

Transport: California ACS 2017, admitted on provenance and schema alone, with a label-blind household hash partition fixed before any prediction existed. **Mode A** applies everything frozen: 2018 services, channels, probes, attacker candidate sets and 2018 validation selections are scored once on 2017. **Mode B** refits only probes, priors and attackers on 2017 fitting pools and selects on 2017 validation pools; representations, protection matrices and nuisances are never refitted. The two are never pooled: a weaker frozen attack after a shift is not evidence that fresh attackers fail.

Attacks are logistic regression, two restarted MLPs at 120/360 epochs, two gradient-boosted trees and a random-feature kernel-ridge family, plus every legal projection of H’s own attacks and of singleton candidates into the coalition view, plus, for the neural arms, their public auxiliary heads and saved-observer catch-up trajectories. Before the final partition could be read, a lock hashed 17,639 inputs—protocol, comparison families, code, all interface objects, all 2017 fitting/validation releases, every fitted object and selection—and was committed and pushed. Uncertainty is a paired household-cluster bootstrap (2,000 replicates over the 10,701 final households) with single-step max- $|t|$ simultaneous intervals inside each pre-declared family.

Data independence. This is transport between public survey years of the same state and age band, not a fresh sample from the 2018 distribution. Public identifiers *cannot* establish that 2017 respondents are different people from 2018 respondents. Verified disjointness is limited to published household groups within 2017. Income is nominal and unadjusted, and the annual adjustment factor differs between years, which limits cross-year interpretation of the income column.

What was registered, and what was not. The protocol, the four comparison families, every endpoint, the uncertainty procedure and all decision rules were fixed, hashed and pushed before the final partition could be read. *No directional prediction was registered:* we did not record in advance which way the primary comparison would go. The result below is therefore a prespecified-rule confirmation on a sealed evaluation, not a called shot. It is also not merely exploratory: fixing the contrasts, the endpoint families, the multiplicity correction and the stopping point before the outcome is what makes the simultaneous intervals mean what they say.

7 Results

7.1 Residence capability, and its limits

Every augmented interface clears the 0.01-nat residence reference over H in every seed, weighting and mode; the spectral arms carry 0.025–0.030 nats against 0.017–0.019 for the frozen neural channel J. Table 5 adds the qualification that matters: the stronger half-headroom reference—is the channel’s residence probe within half the gap between the published service and an *unrestricted* PCA32 or richer bank—passes in only 0–2 of 3 seeds for every interface in both modes. The channel adds real capability relative to what is already published. It does not approach what an unrestricted representation of the same covariates would give.

7.2 Coalition conditioning beats both local controls

On the sealed 2017 partition, C1 has a simultaneous-interval advantage over both its equal-strength local control L1 and the equal-total-mass control L2, under both weightings: 8 of 8 sensitive endpoints better, 0

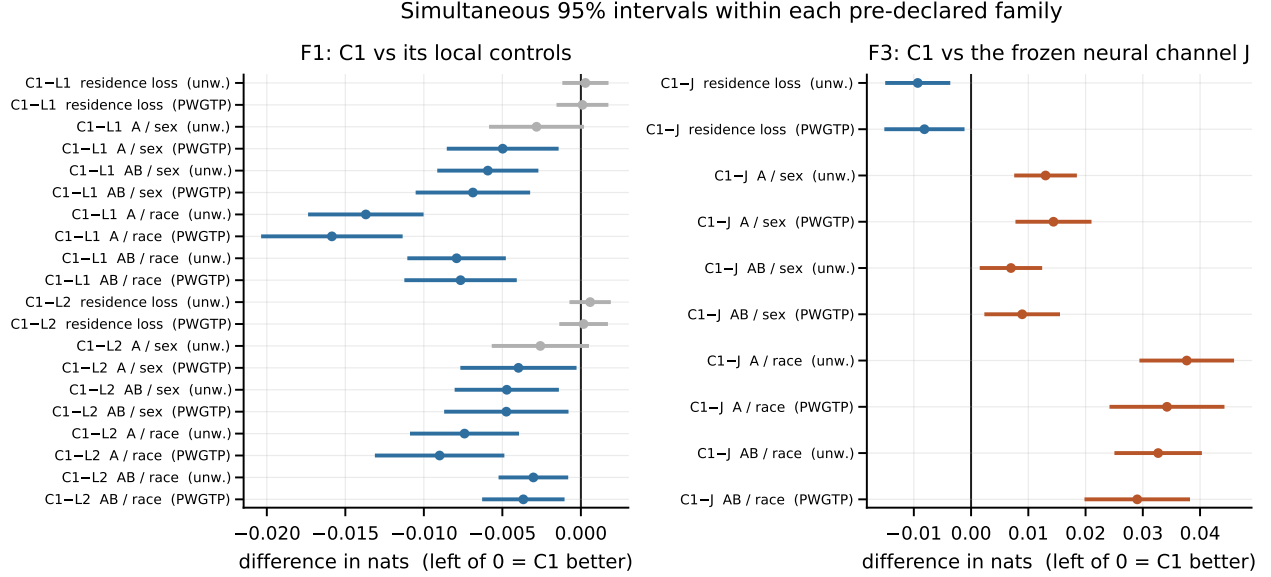


Figure 1: Paired differences with simultaneous 95% intervals inside each pre-declared family, Mode B (fresh 2017 attacks), unweighted and PWGTP. Each difference is candidate minus comparator in nats; **left of zero means C1 is better** for every endpoint shown, including residence loss. Blue: the interval lies entirely below zero. Orange: entirely above zero. Grey: it crosses zero, which means the comparison is unresolved, *not* that the endpoint is unaffected.

worse, critical value 2.9301 over 20 endpoints. Frozen 2018 attacks reproduce the ordering, so it is not an artifact of refitting attackers on the new year. This is the coordination criterion development had failed, and the registered rule passed.

7.3 The residence rule is a point criterion, and only a point criterion

The rule as implemented is $\text{seed-mean}(C1 - \text{comparator}) \leq 0.001$ on the residence log loss. It is one-sided—C1 may be arbitrarily better—and it is a *point* criterion: no interval enters it. The four values are 0.00029, 0.00010, 0.00059, 0.00017, so it passes, and would pass under a two-sided reading too.

It does not follow that the residence cost is below 0.001. Table 2 shows that **zero of four comparisons support non-inferiority at that margin**, missing it by 2% to 35%, and zero support two-sided equivalence. Our earlier phrasing “inside the prespecified 0.001 band” invited exactly the interval reading the evidence does not support, and we withdraw it (Section 8).

A related distinction: the advantage rule requires that *no* sensitive endpoint is significantly worse. That is an absence-of-evidence condition. Two F1 endpoints have adjusted intervals that cross zero on the harmful side—C1 versus L1 on A/sex, $[-0.00585, +0.00020]$, and C1 versus L2 on A/sex, $[-0.00568, +0.00052]$ —so up to 5×10^{-4} nats of *extra* local sex recovery is not excluded. The rule is satisfied; the endpoints are not shown to be unaffected.

7.4 Absolute against additional recovery

The H baseline is substantial: an attacker on the bare service wire already recovers 0.0165 nats of A/race and 0.0141 of AB/sex. Two consequences.

Ratios are scale-dependent; differences are not. C1 against J on A/race is $6.0\times$ on increments (0.0453 vs 0.0076) and $2.6\times$ on absolute values (0.0617 vs 0.0241). The paired difference, 0.0377 [0.0294, 0.0459], is identical on both scales, because the shared prior and H selection cancel inside a paired contrast. We quote the difference.

Some increments are negative, and that is a measurement, not protection. 22 of 429 unweighted role-cells are negative in the primary scope. J’s own A/sex increment is negative under PWGTP (-0.0033 , interval

Contrast	Endpoint	Estimate	Seeds 0 / 1 / 2	Adjusted 95%
<i>F1: coalition conditioning vs local controls ($c = 2.930$)</i>				
C1 – L1	residence loss	0.00029	0.0005 / 0.0004 / -0.0001	[-0.00118, 0.00176]
C1 – L1	A / sex	-0.00282	-0.0046 / -0.0055 / 0.0017	[-0.00585, 0.00020]
C1 – L1	AB / sex	-0.00593	-0.0046 / -0.0094 / -0.0038	[-0.00915, -0.00271] ▼
C1 – L1	A / race	-0.01369	-0.0145 / -0.0129 / -0.0137	[-0.01737, -0.01002] ▼
C1 – L1	AB / race	-0.00792	-0.0054 / -0.0109 / -0.0075	[-0.01105, -0.00478] ▼
C1 – L2	residence loss	0.00059	0.0012 / 0.0009 / -0.0004	[-0.00073, 0.00191]
C1 – L2	A / sex	-0.00258	-0.0020 / -0.0051 / -0.0007	[-0.00568, 0.00052]
C1 – L2	AB / sex	-0.00472	-0.0058 / -0.0060 / -0.0024	[-0.00804, -0.00140] ▼
C1 – L2	A / race	-0.00741	-0.0054 / -0.0074 / -0.0095	[-0.01088, -0.00394] ▼
C1 – L2	AB / race	-0.00302	0.0004 / -0.0039 / -0.0056	[-0.00524, -0.00081] ▼
<i>F3: coalition channel vs frozen neural channel ($c = 2.728$)</i>				
C1 – J	residence loss	-0.00932	-0.0141 / -0.0020 / -0.0118	[-0.01500, -0.00363] ▼
C1 – J	A / sex	0.01301	0.0201 / 0.0034 / 0.0155	[0.00753, 0.01848] ▲
C1 – J	AB / sex	0.00697	0.0088 / 0.0014 / 0.0107	[0.00151, 0.01242] ▲
C1 – J	A / race	0.03767	0.0428 / 0.0383 / 0.0320	[0.02941, 0.04594] ▲
C1 – J	AB / race	0.03268	0.0330 / 0.0318 / 0.0332	[0.02504, 0.04032] ▲

Table 1: Primary comparisons, Mode B, scope `transport_all`, budget 360, unweighted. ▼ marks an adjusted interval entirely below zero (candidate better), ▲ entirely above (worse). The PWGTP rows are in Figure 1.

Contrast	Weight	Estimate	SE	Pointwise 95% upper bound	Simultaneous 95% upper bound	Non-inferior? (margin 0.001)
C1 - L1	unw.	0.00029	0.00050	0.00112	0.00166	no
C1 - L1	PWGTP	0.00010	0.00056	0.00103	0.00164	no
C1 - L2	unw.	0.00059	0.00045	0.00135	0.00182	no
C1 - L2	PWGTP	0.00017	0.00053	0.00102	0.00162	no

Table 2: Retrospective non-inferiority for the F1 residence endpoint at the registered margin of 0.001 nats. Lower loss is better, so the question is whether the *upper* bound falls below +0.001. It does not, in any of the four comparisons, even under the most permissive procedure (a pointwise one-sided bootstrap percentile with no multiplicity adjustment). This assessment is retrospective and does not replace the registered outcome.

excluding zero) and its absolute A/sex recovery, 0.0092, is *below* H’s own 0.0101: the validation-selected fresh attack on J’s wire does worse than H’s attack. That is a property of selection. H’s attack, routed onto J’s augmented wire, remains executable, so a negative increment removes nothing an H-only attacker already had. In 8 of 312 sensitive cells the selected attack generalises worse than that routed H control (15 of 312 under the fresh-only scope). Adding frozen 2018 candidates lowers additional recovery for every interface because it strengthens the subtracted baseline, not because the wire discloses less.

7.5 The mechanism is not competitive

C1 is significantly worse than J on *all four* sensitive endpoints under *both* weightings, and worse on both source probes (income +0.0081, employment +0.0135, intervals excluding zero). It is better only on residence, −0.0093 [−0.0150, −0.0036]. Figure 1, right panel. This is the pre-declared secondary comparison and it fails; our earlier summary said it failed on local race, which understated it.

“Same width” is a *capacity* control—both channels are 16-dimensional—and says nothing about whether their utility is comparable. It is not.

Absolute recovery splits into what H alone already gives and what the channel adds

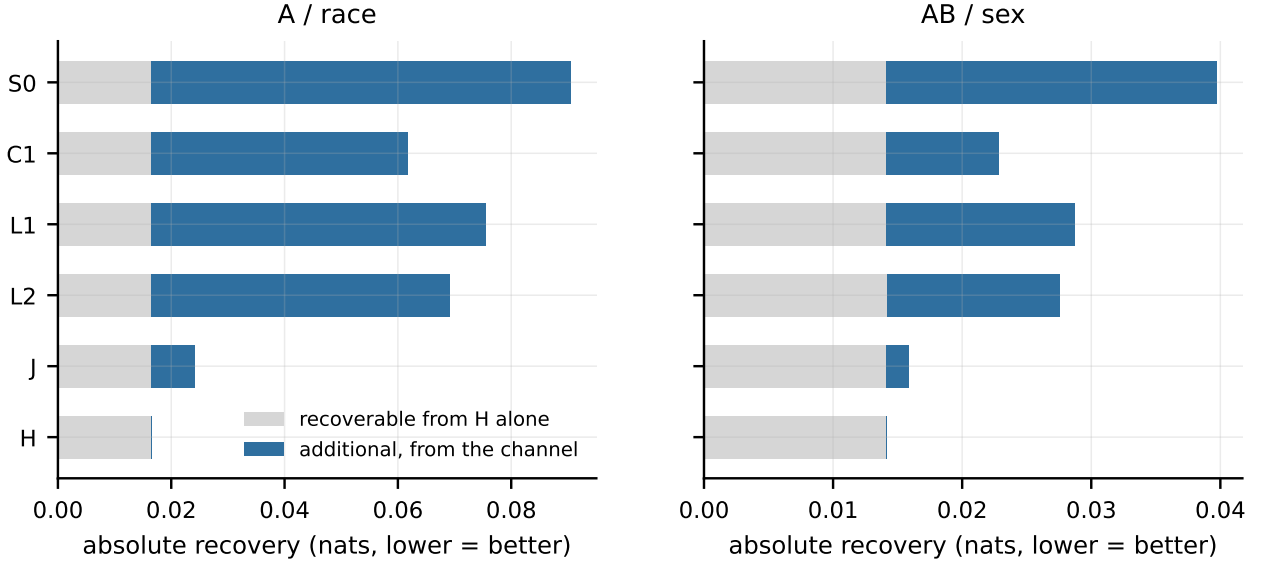


Figure 2: Absolute recovery on the 2017 final partition, split into the part an attacker already gets from the published service alone (grey) and the part the appended channel adds (blue). Mode B, unweighted, seed mean. Lower is better. The grey segment is identical across interfaces by construction: it is H’s own recovery.

Interface	residence gain vs H	A / race		AB / sex	
		absolute	additional	absolute	additional
H	0.0000	0.0165	0.0000	0.0141	0.0000
J	0.0189	0.0241	0.0076	0.0159	0.0017
L1	0.0285	0.0754	0.0590	0.0288	0.0146
L2	0.0288	0.0692	0.0527	0.0275	0.0134
C1	0.0282	0.0617	0.0453	0.0228	0.0087
S0	0.0304	0.0905	0.0740	0.0397	0.0256

Table 3: Absolute and additional (H-relative) recovery beside residence gain. Mode B, `transport_all`, 360, unweighted, seed mean. Lower is better for recovery, higher for residence gain.

7.6 Matching the two channels on average utility

Because the routed family mixes per-person *losses*, expected residence gain over H is exactly p times the full-release gain, and expected additional recovery is exactly p times the full-release increment. The probability matching a target’s *mean* residence gain is therefore identified in closed form. This yields two findings the fixed- p grid did not surface.

First, **J cannot be made as useful as C1 by any withholding schedule**: it would need $p^* = 1.49$. Nor can E (1.07), L025 (1.62) or L20 (1.56). The C1-versus-J comparison is not two points on one traversable frontier. Second, **the ordering survives matching in both directions**. Where matching is identifiable, C1 beats withheld A0 ($p^* = 0.976$) and withheld S0 ($p^* = 0.929$) on all four sensitive endpoints; and C1 withheld down to J’s residence level ($p^* = 0.670$) still leaks far more than J on all four (0.0303 against 0.0076 on A/race).

Both statements are **post-hoc average-utility matching**: p^* is estimated from the same data as the outcome, so the fixed- p intervals do not transfer to it, and it matches an average over people who receive different information. The original fixed- p analysis is unchanged, and on it neither family robustly dominates: withholding dominates a given spectral arm in 1–3 of 180 fixed comparisons and a spectral arm dominates withholding in 0–3 of 180. A finite list of controls failing to dominate C1 is not evidence that C1 is efficient.

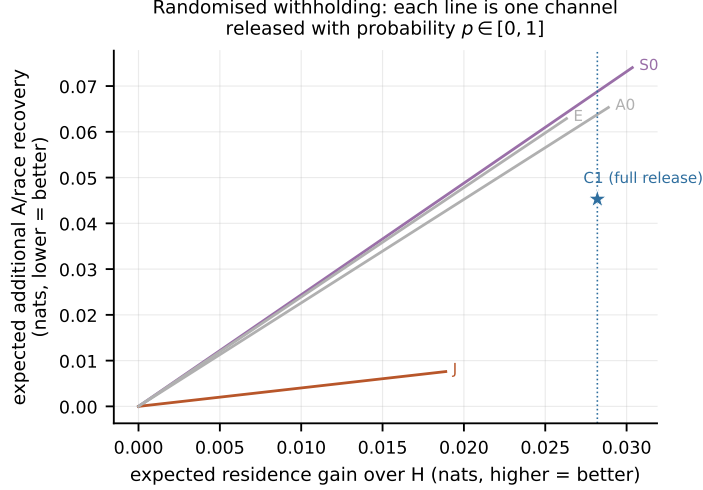


Figure 3: Under the declared randomised-withholding mechanism (H always released; an independent visible Bernoulli(p) branch, fixed per person, decides whether the channel is also released), expected residence gain and expected additional recovery are both exactly p times their full-release values, so each channel traces a straight line from the origin. Lower and further right is better. The dotted line marks C1’s full-release residence gain: **J’s line stops short of it**, so no withholding schedule makes J as useful as C1. Where a comparator *can* reach that line (S0, A0), it sits above C1.

Withheld channel	matched to	p^*	identifiable?	A/race at p^*
E	C1	1.072	no	—
A0	C1	0.976	yes	0.0638
L025	C1	1.615	no	—
L20	C1	1.555	no	—
J	C1	1.493	no	—
S0	C1	0.929	yes	0.0688

Table 4: Post-hoc average-utility matching. $p^* = \text{gain}_{C1} / \text{gain}_{\text{source}}$ is the withholding probability at which a channel’s *mean* residence gain equals C1’s; it is identified in closed form because the mixture is exactly linear in p . Values above 1 mean the comparator cannot reach C1’s utility at any schedule. The last column is the comparator’s expected additional A/race recovery at p^* , to compare against C1’s 0.0453.

7.7 Service parity, probe quality, and what moved

Three different objects must be kept apart. The *released* income, employment and coverage probability vectors are identical, bitwise, in every condition—structural, verified on 210 of 210 released views. Service *accuracy* changes with the year: employment +0.0138 nats, income −0.0145, coverage −0.0067. Independently fitted *source probes* are a third thing again, and their pass pattern separates the two modes (Table 5).

7.8 Development against transport: magnitude, not only resolution

All 20 F1 *seed-mean* signs agree with development. Only 13 of 20 agree in *every seed* (Figure 4); across all 90 family endpoints only 48 of 90 have all three seeds agreeing with their own seed mean. And magnitudes moved: F1 point estimates changed by factors between 0.054× and 14.8× (for instance C1 – L2 on AB/sex went from −0.0003 to −0.0047). A larger evaluation pool shrinks standard errors; it does not move point estimates by a factor of fifteen. The years differ in sample size, in calendar year, in evaluation pool and in which attacker families were available, and this design does not separate them. Our earlier claim that “what changed is resolution, not direction” is withdrawn.

Interface	source-probe allowance		half-headroom	
	fresh (B)	frozen (A)	fresh (B)	frozen (A)
H	3/3	3/3	0/3	0/3
J	3/3	3/3	0/3	0/3
L1	3/3	0/3	1/3	0/3
L2	3/3	0/3	1/3	0/3
C1	3/3	0/3	1/3	0/3
S0	3/3	0/3	1/3	2/3

Table 5: Passing seeds out of 3 for the two reference criteria, unweighted. The legacy source-probe allowance passes everywhere under fresh adaptation (B) but **passes in only 0-1 of 3 seeds for every spectral arm under frozen transfer (A)**: a frozen 2018 probe does not carry onto a spectral wire in a new year, and refitting the probe repairs it. The released service vectors are identical bits in all cases; a worse probe does not change a delivered prediction.

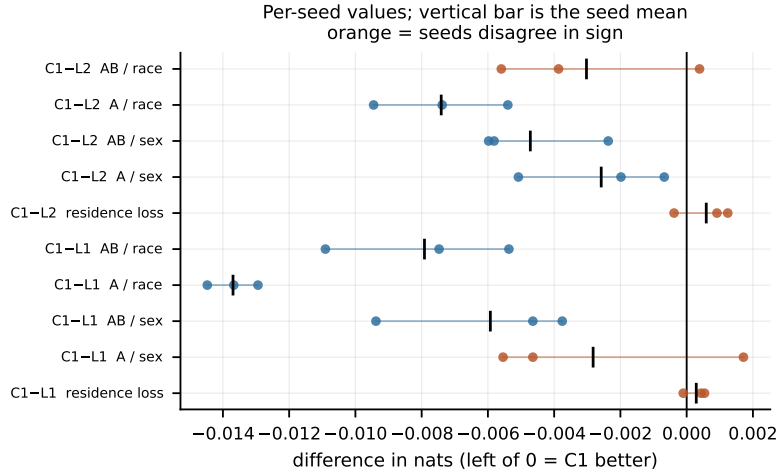


Figure 4: The three per-seed values behind each F1 endpoint (unweighted); the vertical bar is the seed mean that enters the estimand. Orange marks endpoints where the seeds do not all share the seed-mean sign. Aggregate sign agreement can conceal per-seed disagreement, and does here.

7.9 Support limitations

Race class 3 (Alaska Native alone) has one person in the final partition and none in 2017 attacker validation, so class-specific recall and AUROC are undefined for it. Full nine-class log loss is always scored with a fixed 10^{-12} probability floor and renormalisation; no category is dropped. A comparable-category diagnostic restricted to classes supported in every fitting pool of both modes reproduces the F1 race differences almost exactly (C1 – L1 on A/race: -0.0137 against -0.0137 on all rows). **A class without support is unmeasured, not protected.**

8 Claims this revision withdraws or narrows

Stated explicitly, because each was in our own earlier draft.

1. “*Residence difference inside the prespecified ± 0.001 band.*” The rule is one-sided and applies to a point estimate. Zero of four interval comparisons support non-inferiority at that margin (Table 2). The registered rule still passed.
2. “*What changed is resolution, not direction.*” Magnitudes changed by $0.054\times$ to $14.8\times$; sample size alone does not explain the newly significant result.
3. “*Every F1 endpoint keeps its development sign (20/20).*” True of seed means; 13 of 20 in every seed.

4. “*C1 versus J fails on local race.*” It fails on all four sensitive endpoints under both weightings, plus both source probes.
5. “*Several times more sex and race.*” Scale-dependent: $6.0\times$ on increments, $2.6\times$ in absolute terms. We now quote the paired difference.
6. “*Under frozen transfer every spectral arm fails in 0–1 of 3 seeds.*” The underlying column counts *passing* seeds; the arms pass in 0–1 of 3, i.e. fail in 2–3 of 3. The published table was correct and the sentence inverted it.
7. “*The gap is the surrogate, not the solver.*” Global optimality of the fixed matrix excludes a solver failure and nothing else. Five candidate causes—the finite feature map, nuisance misspecification, the normalisation, the fixed rank, the reconstruction surrogate, and finite-sample generalisation—remain unseparated. We now say “consistent with”.
8. “*If a nonlinear penalty also loses, the conclusion generalises to the closed-form spectral family.*” Withdrawn entirely. Two arms failing is evidence about two arms; and a penalty acting on nonlinear functions of Z is provably not a trace form $\text{tr}(W^\top AW)$ for any W -independent A —it is not invariant under $W \mapsto WQ$ for orthogonal Q , while every trace form is—so such a successor is not in that family at all.
9. “*Two lock amendments.*” Three exist. All are code-only; the third post-dates every final-partition read.
10. *Reporting the H-relative residence gain without the half-headroom failure.* Corrected in Table 5.

9 Limitations

Three seeds are three fixed fitted systems on one cohort, not population replications. The bootstrap covers people in the 2017 final partition only: it is not retraining variability and not a Census-design variance, and PWGTP replicate weights were not used. Attack strength is a floor, not a guarantee: every recovery number is what the tested families achieved. Only recipient A receives a channel. Transport is between public survey years of the same state and age band, and cross-year person distinctness cannot be established from public identifiers. Income is nominal and unadjusted. The evaluation refits attackers but not the representation, so it does not test a mechanism retrained on the new year. The 2018 pools have been used repeatedly across this line of work and are development, not confirmation.

Numerical recovery. Three code-only lock amendments are recorded. The first fixed a release-file write race; the second fixed rare load-dependent prediction corruption found by an independent cross-process replay; the third changed three published tables from CSV to gzip CSV and post-dates every final read. Machine memory pressure was the *observed condition* under which the corruption appeared; it is not a demonstrated cause, and we do not claim one. The repair does not depend on the diagnosis: every prediction is now computed twice and must agree, all final units were rescored, originals are preserved, and the post-fix replay found 0 mismatches in 59,874 predictions with 0 selection changes in 37,122 validation predictions.

10 Conclusion

A service that has already published predictions can add a reusable channel beside them without changing a single released number, and can steer that channel away from directions a recipient or a colluding pair would use to infer sex and race. On a locked, previously unused survey year, the coalition-conditioning knob does what it claims against both of its local controls, under simultaneous intervals and both weightings. It also costs residence utility by an amount the data cannot bound below the 0.001 nats our registered rule used, and the channel it steers remains significantly worse than a frozen adversarially trained channel of the same width on every sensitive endpoint—an ordering that survives matching the two on average utility in both directions.

The defensible contribution is the problem formulation and the evaluation design: fixed published outputs as a hard constraint, recipient-specific and coalition-specific audit roles, fresh and frozen attackers never pooled, absolute and incremental recovery reported side by side, negative increments retained unclipped, and a sealed temporal transport with the multiplicity fixed in advance. The method result is negative and we report it as such.

11 Reproducibility

Deterministic seeds throughout; every unit writes a hash-bound completion record; the lock refuses to open the final partition if any hashed input changed. All 90 pre-declared endpoints in this paper were regenerated from stored per-person predictions by an independently written scorer that shares no computation function with the original report code, agreeing to 1.1×10^{-15} on point estimates and 2.4×10^{-15} on interval endpoints, with identical critical values; candidate selection was re-derived from stored validation losses with 0 mismatches in 924 cells; and the lock was re-verified over all 17,639 hashed inputs with every amendment applied. Commands, file map and software versions: `REPRODUCE.md`. Fitted objects, releases and per-person arrays stay local; compact aggregates, code, tables and figures are published.

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